Controllable scientific balloon

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Introduction

This capstone project presents a high-altitude balloon system with active control designed for tropospheric missions. A scientific balloon has many uses, including weather monitoring, astronomy, and UAV launching. In all cases, these balloons are not controlled or do not have the capacity to stay at one location. As a consequence, due to high-altitude winds, they drift sometimes hundreds of kilometers. To solve this issue, a control system is considered so that the balloon can navigate through the wind. For this project, lower altitude ballooning is considered (7 km) due to its relatively lower winds.

Governing Equations

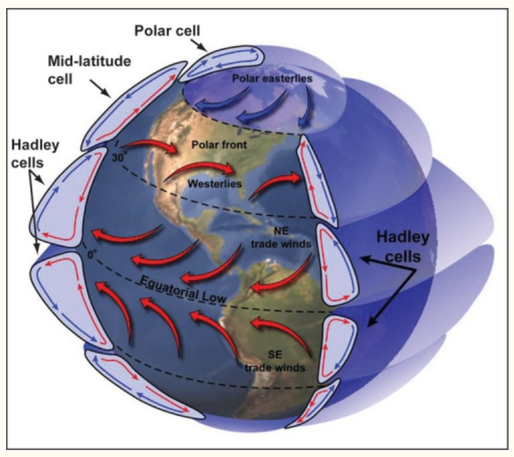
The motion of the balloon is governed by its net force: $\mathbf{F_{net}} = \mathbf{F_B} - \mathbf{F_W} - \mathbf{F_D}$

-Buoyant Force: $F_B = \rho_a \times V \times g$
-Gravitational Force: $F_W = m_t \times g$
-Drag Force: $F_D = 0.5 \times \rho_a \times C_D \times A_B \times \mathbf{u} \times |\mathbf{u}|$

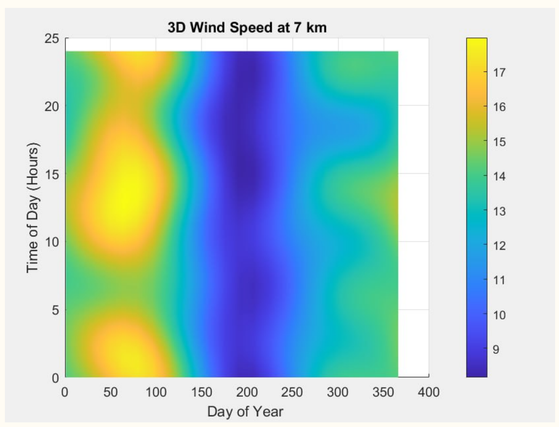
For determining the angle of the gondola, torque is used: $\boldsymbol{\tau} = \mathbf{I} \times \boldsymbol{\alpha}$. These equations define the dynamic response of the balloon to environmental and control forces.

Troposphere and Wind Analysis

Wind velocity increases with altitude in the troposphere. The wind at high altitude has low variation, due to lesser effects of friction winds, which are more chaotic. MATLAB was used to analyze wind speed at different altitudes and determine the disturbance profile affecting the balloon.

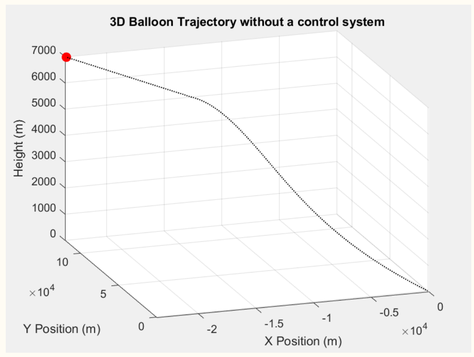


The wind circulation in the troposphere.



This heatmap shows wind speed variation at 7 km throughout the year and different times of day. High speeds (yellow) occur early in the year, while calmer conditions (blue) appear mid-year.

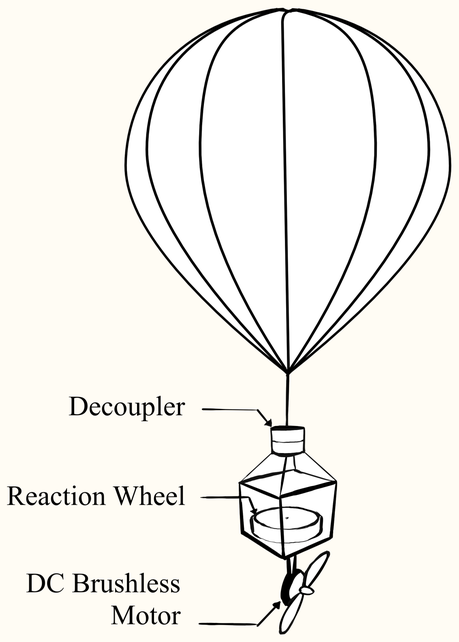
Control Method



This 3D graph shows the balloon drift over the course of 5.5 hours without any control system. It drifts more than 100km within that time. For this reason, a control system was devised.

The system which most correctly tackles this issue uses two controls:

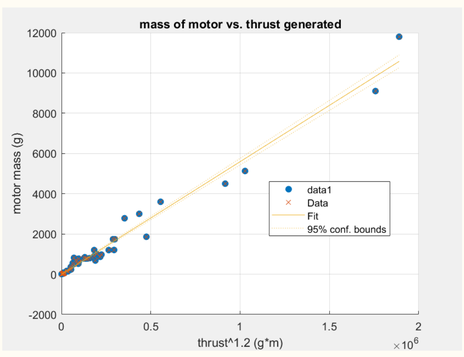
1. Directional control with a brushless DC motor fixed beneath the gondola.
2. Angular control using a reaction wheel for steering the gondola's angle and a decoupler, which separates the gondola from the balloon.



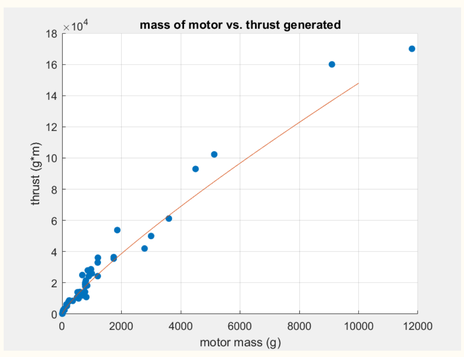
Sketch of the control system for the balloon

Thrust-to-Mass Motor Linear Regression

One of the challenges that comes with building such a system is that the stronger the motor is, the more massive it will be. To assess this relationship, existing motors with varying thrusts were taken and a linear regression was done. The relationship is not perfectly linear, so a linearization was done. The approximate equation to relate mass to thrust is $m = 0.006235 \times T^{1.2}$, where mass is in grams and thrust is in gram.meters.



The linearized data and its linear regression.



Data with the linear regression equation.

Three Variable Optimization

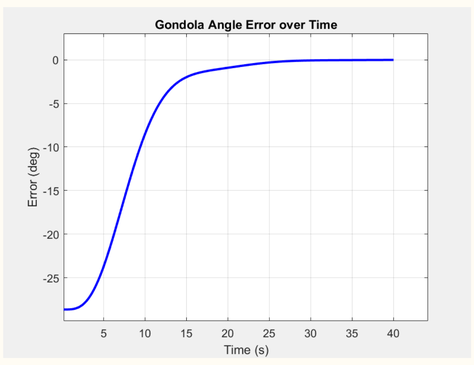
To accurately assess the necessary parameters which may allow the balloon to function properly, a three variable optimization was conducted. Three variables were considered: volume, mass and thrust. These variables are interconnected such that if one is changed, others will also need to change. The volume of the balloon is dependent on the total mass of the system. The thrust needed to withstand the drag force is dependent on the balloon's volume, and the total mass is dependent on the thrust of the motor.

Iteration	Motor mass (g)	Volume (m^3)	Thrust (kg.m)
1	500	2.324755974	28.31111892
2	1371.465536	3.844214507	39.58924274
3	2050.827307	5.028727258	47.35263746
4	2542.432702	5.885874192	52.5911623
5	2883.578513	6.480684764	56.07716509
6	3114.438205	6.8832041	58.3757604
7	3268.251872	7.151388531	59.88237752
8	3369.731515	7.328325087	60.86607126
9	3436.266055	7.444332512	61.50672852
10	3479.714488	7.520087766	61.92329515
11	3508.014082	7.569430004	62.19386844
12	3526.415994	7.60151497	62.36949427

Example of the iterations for the three variable optimization. The output of one function is the input for the next one in a cyclical manner. If the system of equations converge, then such a value exists and a relative error is used as the stopping condition.

Angle Control

An LQR (Linear Quadratic Regulator) controller was designed to minimize angular deviation and regulate the performance to effort ratio effectively. The signal is transmitted to the decoder and the reaction wheel, which spin and create a gyroscopic effect.



The gondola's angle error is found.



The reaction wheel's speed is found. Through the manipulation of the Q and R matrices in the LQR system, it is possible to adjust the effort to get less performance if battery needs to be preserved.

Acknowledgment

We sincerely thank our supervisor, Dr. Hrachya Kocharyan, for his guidance and support throughout this project.

Conclusion

The project demonstrates the feasibility of navigating with the balloon. The control design of the balloon is working within the required parameters. For higher altitudes, this design will not work due to the increasingly high winds at higher altitudes. Nevertheless, future work is necessary to find the optimal path the balloon should take to reach the target point with angle and speed control as well as GPS implementation with error minimization.